

DSAA3073: Theories in Data Science

Week 1 Validation Sheet: Introduction & Mathematical Preliminaries

Total Points: 130

Instructions:

- Write your answers clearly and legibly.
- Show all working for full credit.
- State any assumptions you make.
- For proofs, justify each step rigorously.
- This is a practice validation sheet. AI tools and references are allowed.
- Use this to assess your understanding before the actual test.

Part A: Multiple Choice (40 points)

Each question is worth 5 points. Select the single best answer.

Question 1. [5 marks] Which of the following statements about Markov's inequality is TRUE?

- A) Markov's inequality can be applied to any random variable, including those taking negative values.
- B) Markov's inequality gives tighter bounds than Chebyshev's inequality when the variance is small.
- C) For a non-negative random variable X with $\mathbb{E}[X] = \mu$, Markov's inequality states that $\mathbb{P}[X \geq a] \leq \frac{\mu}{a}$ for any $a > 0$.
- D) Markov's inequality requires the random variable to be bounded in a known interval.

Question 2. [5 marks] Let $X_1, \dots, X_n$ be i.i.d. random variables with $X_i \in [0, 1]$ and $\mathbb{E}[X_i] = \mu$. Which statement about the convergence of $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i$ to μ is CORRECT?

- A) The Weak Law of Large Numbers guarantees that $\hat{\mu} = \mu$ for sufficiently large n .
- B) Hoeffding's inequality provides a finite-sample bound on $\mathbb{P}[|\hat{\mu} - \mu| > \varepsilon]$ that decays exponentially with n .
- C) Chebyshev's inequality guarantees faster convergence than Hoeffding's inequality for bounded random variables.
- D) The union bound is needed to prove the convergence of $\hat{\mu}$ to μ .

Question 3. [5 marks] Consider norms in $\mathbb{R}^d$. Which of the following relationships is TRUE for all $x \in \mathbb{R}^d$?

- A) $\|x\|_1 \leq \|x\|_2 \leq \|x\|_\infty$
- B) $\|x\|_\infty \leq \|x\|_1 \leq \sqrt{d}\|x\|_\infty$
- C) $\|x\|_\infty \leq \|x\|_2 \leq \sqrt{d}\|x\|_\infty$
- D) $\|x\|_2 \leq \|x\|_1 \leq d\|x\|_2$

Question 4. [5 marks] Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be twice-differentiable. Which condition is NECESSARY for f to be convex?

- A) $f''(x) > 0$ for all x (strictly positive second derivative).
- B) $f''(x) \geq 0$ for all x (non-negative second derivative).
- C) f has a unique global minimum.
- D) $f(x) \geq 0$ for all x .

Question 5. [5 marks] The triangle inequality states that $\|x + y\| \leq \|x\| + \|y\|$. When does equality hold for the ℓ_2 norm?

- A) When x and y are orthogonal.
- B) When x and y point in the same direction (i.e., $y = \alpha x$ for some $\alpha \geq 0$).
- C) When $\|x\| = \|y\|$.
- D) When x and y are unit vectors.

Question 6. [5 marks] Consider a finite hypothesis class $\mathcal{H}$ with $|\mathcal{H}| = m$. You want to ensure that with probability $\geq 1 - \delta$, all hypotheses in $\mathcal{H}$ have empirical risk within ε of true risk. Using Hoeffding's inequality and the union bound, what is the required sample size n ?

- A) $n \geq \frac{\ln(\frac{2}{\delta})}{2\varepsilon^2}$
- B) $n \geq \frac{\ln(2\frac{m}{\delta})}{2\varepsilon^2}$
- C) $n \geq \frac{\ln(\frac{m}{\delta})}{\varepsilon^2}$
- D) $n \geq \frac{m \ln(\frac{2}{\delta})}{2\varepsilon^2}$

Question 7. [5 marks] Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a convex function. Which statement is TRUE?

- A) If $\nabla f(x) = 0$, then x is a local minimum but not necessarily a global minimum.
- B) Any local minimum of f is also a global minimum.
- C) The set of global minima of f must be a singleton (single point).
- D) Convex functions must be differentiable everywhere.

Question 8. [5 marks] Which of the following best describes the purpose of concentration inequalities in learning theory?

- A) They prove that learning algorithms converge to the optimal hypothesis.
- B) They provide finite-sample guarantees that the empirical average is close to the true expectation with high probability.
- C) They show that the VC dimension of a hypothesis class is finite.
- D) They guarantee that gradient descent finds the global minimum.

Part B: Mixed Format Questions (90 points)

Question 9. [12 marks] Let X be a random variable with $\mathbb{E}[X] = 5$ and $\text{Var}(X) = 4$.

- (a) [4 marks] Use Chebyshev's inequality to bound $\mathbb{P}[|X - 5| \geq 6]$.
- (b) [4 marks] If additionally X is non-negative, use Markov's inequality to bound $\mathbb{P}[X \geq 15]$.
- (c) [4 marks] Compare the bounds from parts (a) and (b). Which provides more information, and why?

Question 10. [15 marks] Consider vectors $x = (3, 4)$ and $y = (5, 12)$ in $\mathbb{R}^2$.

- (a) [3 marks] Compute $\langle x, y \rangle$, $\|x\|$, and $\|y\|$ where $\|\cdot\|$ denotes the ℓ_2 norm.
- (b) [6 marks] Verify the Cauchy-Schwarz inequality for these vectors: $|\langle x, y \rangle| \leq \|x\|\|y\|$.
- (c) [6 marks] Compute $\|x + y\|$ and verify the triangle inequality: $\|x + y\| \leq \|x\| + \|y\|$.

Question 11. [13 marks] Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^2$.

- (a) [5 marks] Show that f is convex by verifying the first-order characterization: $f(y) \geq f(x) + f'(x)(y - x)$ for all $x, y \in \mathbb{R}$.
- (b) [4 marks] Verify the second-order condition: $f''(x) \geq 0$ for all x .
- (c) [4 marks] Find all global minima of f and explain why any critical point of a convex function is a global minimum.

Question 12. [15 marks] Consider i.i.d. random variables $X_1, \dots, X_n$ where $X_i \in [a, b]$ with $\mathbb{E}[X_i] = \mu$.

- (a) [6 marks] State Hoeffding's inequality for the sample mean $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i$.
- (b) [5 marks] For the case $[a, b] = [0, 1]$, derive the minimum sample size n needed to guarantee $\mathbb{P}[|\hat{\mu} - \mu| > 0.1] \leq 0.01$.
- (c) [4 marks] Explain why Hoeffding's inequality is more useful in practice than the Weak Law of Large Numbers for determining sample size requirements.

Question 13. [10 marks] Consider the ℓ_1 and ℓ_∞ norms in $\mathbb{R}^d$.

- (a) [4 marks] Prove that $\|x\|_\infty \leq \|x\|_1$ for all $x \in \mathbb{R}^d$.
- (b) [6 marks] Show that $\|x\|_1 \leq d \cdot \|x\|_\infty$ for all $x \in \mathbb{R}^d$. Give an example showing this bound is tight (cannot be improved).

Question 14. [10 marks] Let X be a random variable taking values in $[0, 1]$ with $\mathbb{E}[X] = 0.2$.

- (a) [5 marks] Use Markov's inequality to bound $\mathbb{P}[X \geq 0.6]$.
- (b) [5 marks] Explain why we cannot use Chebyshev's inequality to improve this bound without knowing the variance of X . If we additionally know that $\text{Var}(X) = 0.04$, compute the Chebyshev bound for $\mathbb{P}[X \geq 0.6]$ and compare with the Markov bound.

Question 15. [15 marks] The exponential moment method is a proof technique for deriving concentration inequalities.

- (a) [7 marks] Explain the key idea: why do we transform $\mathbb{P}[X \geq a]$ into $\mathbb{P}[\exp(tX) \geq \exp(ta)]$ for $t > 0$, and how do we then apply Markov's inequality?
- (b) [8 marks] Let $X \in [0, 1]$ with $\mathbb{E}[X] = 0.4$. Assuming $\mathbb{E}[\exp(tX)] \leq \exp\left(t\mathbb{E}[X] + \frac{t^2}{8}\right)$, use the exponential moment method to bound $\mathbb{P}[X \geq 0.9]$. Optimize over t to find the tightest bound.

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